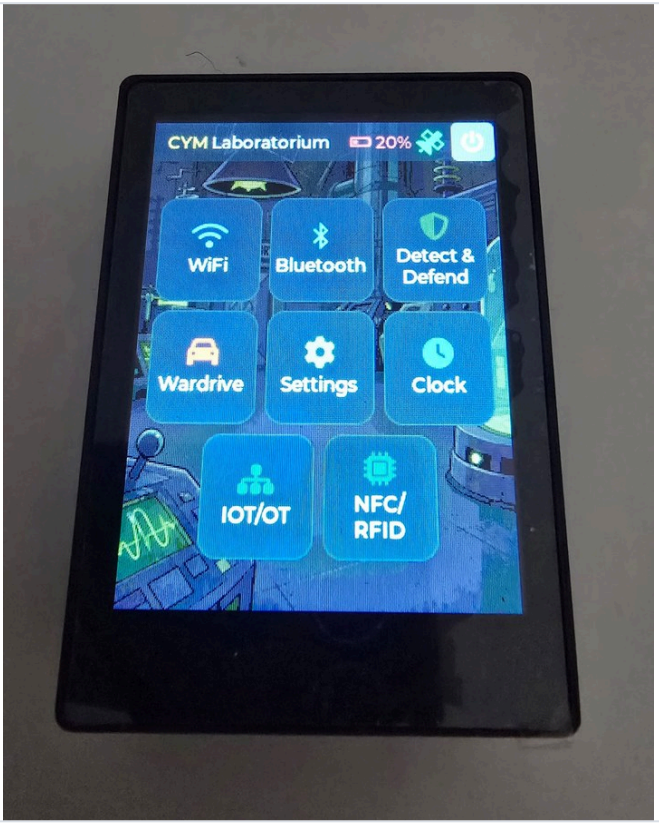
CYM Case WS-C5-28 – 3D Printing and Installation Manual

Case with battery and GPS for the Waveshare ESP32-C5-Touch-LCD-2.8 · Revision 4

Designed and made by @birolt29 @alen29





The real device (CYM menu open) and the 3D design view (front).

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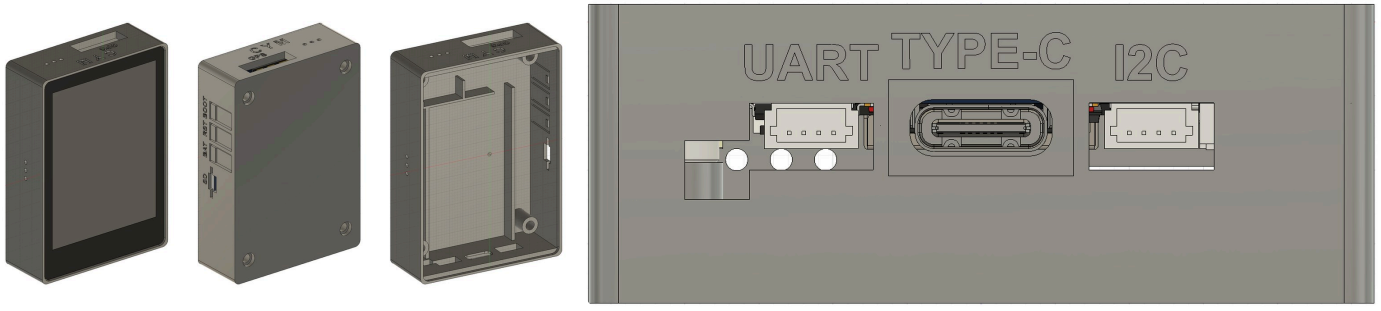
Case with battery and GPS for the Waveshare ESP32-C5-Touch-LCD-2.8 · Revision 4

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This manual has four parts: general information about the board and the parts used, 3D printing of the case, assembling the parts into the case, and installing the firmware. Print settings are given by their generic names; you will find the equivalents in whichever slicer you use.



Finished device: front, boot screen, button wall, USB wall.



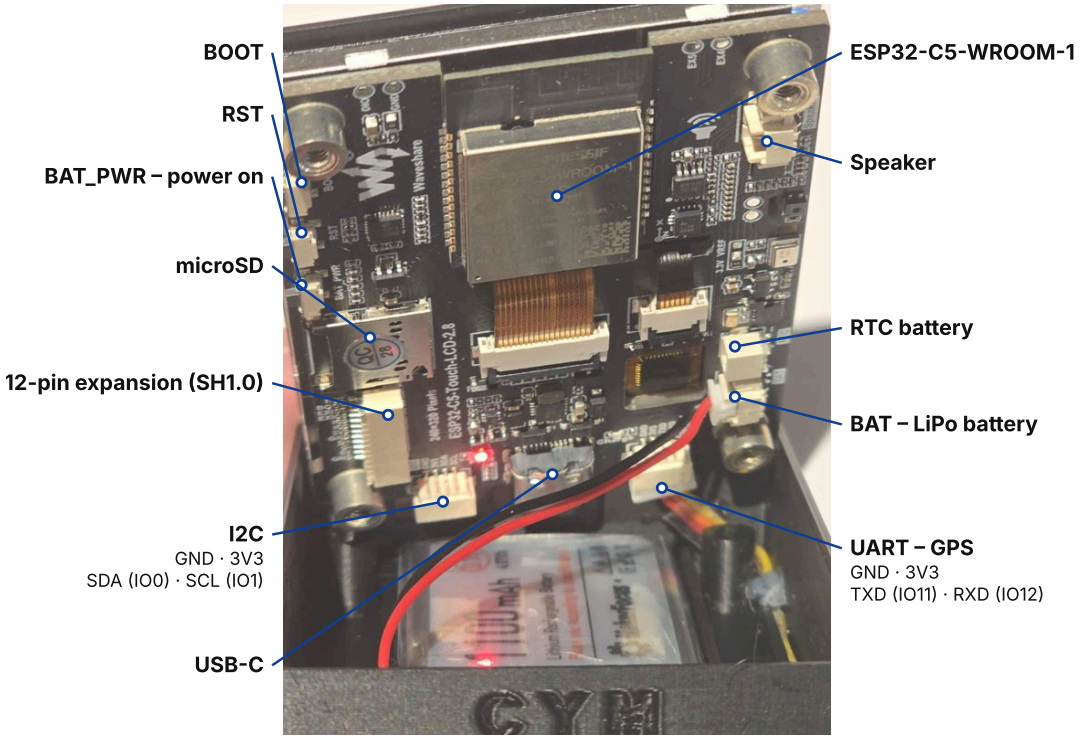
Design views: front, back, inside of the case, USB wall.

Part 1 – General information

Board specifications

Processor	ESP32-C5 · RISC-V single-core · 240 MHz
Memory	32 MB Flash · 8 MB PSRAM
Wi-Fi	802.11 a/b/g/n/ax (Wi-Fi 6) · 2.4 GHz + 5 GHz dual-band
Bluetooth	Bluetooth 5 (LE)
IEEE 802.15.4	Zigbee / Thread support
Display	2.8" IPS · capacitive touch
microSD	FAT32 · max 32 GB
Sockets	USB-C · UART · I2C · BAT (battery)
Buttons	BOOT · RST · BAT (power)
Power	5 V via USB-C · or battery (BAT socket)
Case dimensions	53.0 × 75.5 × 20.2 mm

Connectors (pinout)



Back of the board, USB-C at the bottom. Pin order (top to bottom on each socket, also printed on the board) and GPIO numbers are from the Waveshare documentation. The UART socket supplies 3.3 V.

Reference: parts and firmware used

Part	Product	Link
Board	Waveshare ESP32-C5-Touch-LCD-2.8-EN IPS Touch Display Development Board	www.amazon.de/dp/B0H7R28LZN
Battery	YELUFT 3.7V 1100mAh LiPo Battery for ESP32, Meshtastic Node	www.amazon.de/dp/B0FX8M8DK7
GPS module	ATGM336H GPS+BDS Dual Mode Module	www.amazon.de/dp/B0FKH26344
Firmware	Cheap Yellow Monster (CYM)	github.com/JimGat/CYM#supported-hardware

The case was tested with these parts. Equivalent parts of the same size also fit (battery 53 × 30 × 7.5 mm, GPS module 16 × 14 × 5.5 mm). Links as of 2026-10-02; product pages may change over time. The firmware supports this board under the name WS-C5-28; select this board when installing.

> Before plugging the battery into the board, check the polarity with a multimeter: on batteries with a 1.25 mm plug it may be reversed compared to the BAT socket on the board.

> Battery size: do not trust the size given in listings. The listing for the battery used here says 50 × 29 × 7 mm; its measured size is 53 × 30 × 7.5 mm (including the protection board and the cable exit). The battery pocket of the case is made for this size.

Part 2 – 3D printing

Print files

File	Contents
`CYM_Case_WS-C5-28.3mf`	Print file; includes the unit information. Use this one if your slicer can open 3MF.
`CYM_Case_WS-C5-28.stl`	The same model as STL. Units: millimeters.

The files contain the case only; it is a single part and needs no supports. Outer dimensions 53.0 × 75.5 × 20.2 mm, volume 13.4 cm³ (about 16 g of PLA).



The printed case: back, USB wall, top (antenna) wall, button wall.

Placement on the bed

- The flat, closed base of the case (the face with the four screw holes) goes on the bed.
- The open side and the display frame face up. The file is saved in this orientation; do not rotate it.
- Supports off. No supports are needed anywhere.
- No brim is needed; add 3–5 mm if the corners lift.

Material

PLA or PLA+ is sufficient. PETG also works and makes the three flexible buttons in the side wall more durable. Choose temperatures from your filament's label; because the part is printed slowly, the lower-to-middle part of the range works well. Damp filament causes stringing and fills in the small lettering; dry it before printing.

Slicer settings (0.4 mm nozzle)

Setting	Value	Why
Layer height	0.16 mm	For the small lettering and holes. 0.20 mm also works, with a coarser result.
Walls (perimeters)	3	The side walls are 1.2 mm thick; three lines print them fully solid.
Wall generator	Arachne / variable line width	The thin 0.69 mm and 0.75 mm walls print in one piece.
Wall order	Outer wall first	Sharper outer surface and lettering edges.
Outer wall speed	30 mm/s	For the engraved lettering and the button slits.
Flow ratio	Calibrated; avoid over-extrusion	Over-extrusion closes the 0.6 mm button slits and the lettering.
Top / bottom solid layers	4 / 4	

Infill	15%	The part is thin; most of it prints solid anyway.
Elephant foot compensation	0.10 mm	Keeps first-layer squish from narrowing the screw counterbores.
Supports	Off	

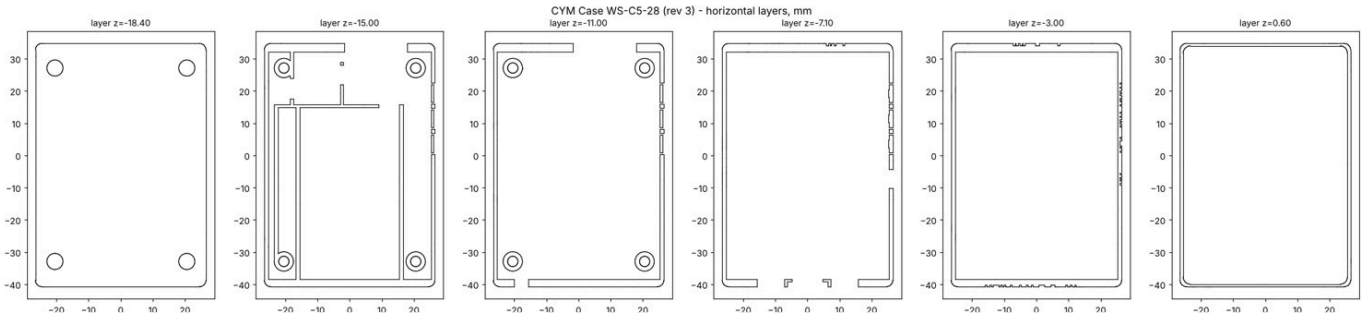
Setting names differ between programs (Cura, PrusaSlicer, OrcaSlicer, Bambu Studio, Creality Print and others). Reference print: an Ender-3 S1 Pro class printer, 0.4 mm nozzle, PLA+, nozzle 210 / 205 °C, bed 60 / 55 °C; about 1 hour 45 minutes.

Check the slicer preview

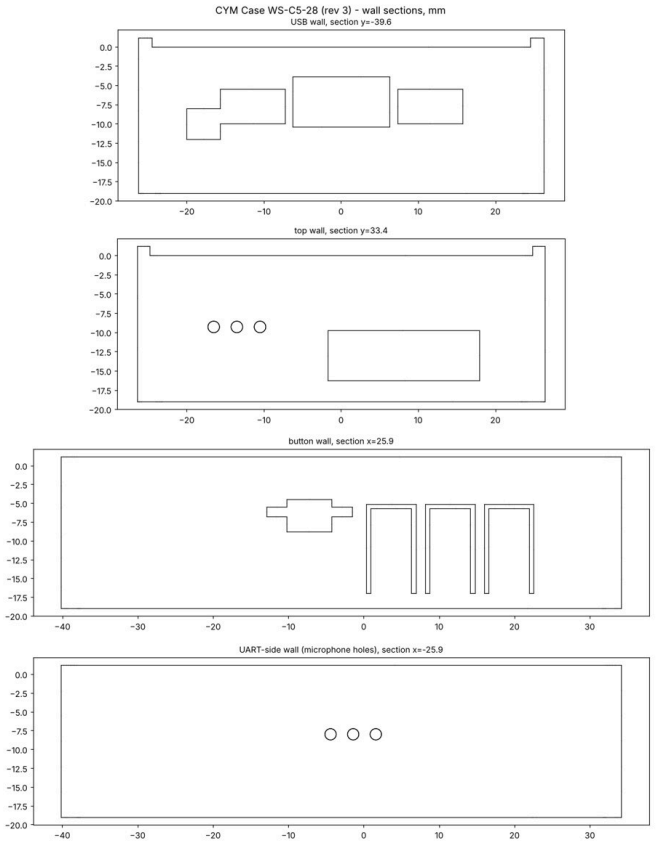
Heights measured from the bed:

Height	What you should see
0 – 1.2 mm	Solid base; only the four screw counterbores (dia. 5 mm) are open.
2.75 – 9.25 mm	GPS antenna opening in the top short wall (19.6 × 6.5 mm).
7.0 – 13.75 mm	UART, USB-C and I2C openings in the USB wall; cable pass-through on the corner side of the UART opening.
9.0 – 11.75 mm	Three GPS LED holes in the top wall, three microphone holes in the long side wall (1.5 mm).
2.0 – 14.5 mm	Three inverted-U slits in the button wall; the SD card opening with the tweezer notch in the middle.
19.0 – 20.2 mm	Frame surrounding the display.

The lettering UART, TYPE-C, I2C, GPS, C Y M, BOOT, RST, BAT and SD must show as 0.6 mm deep grooves in the walls.



Horizontal sections (mm). z = 0 is the top edge of the case; the base is at z = -19.



Wall sections (mm): USB wall, top wall, button wall, long wall on the UART side.

Checks after printing

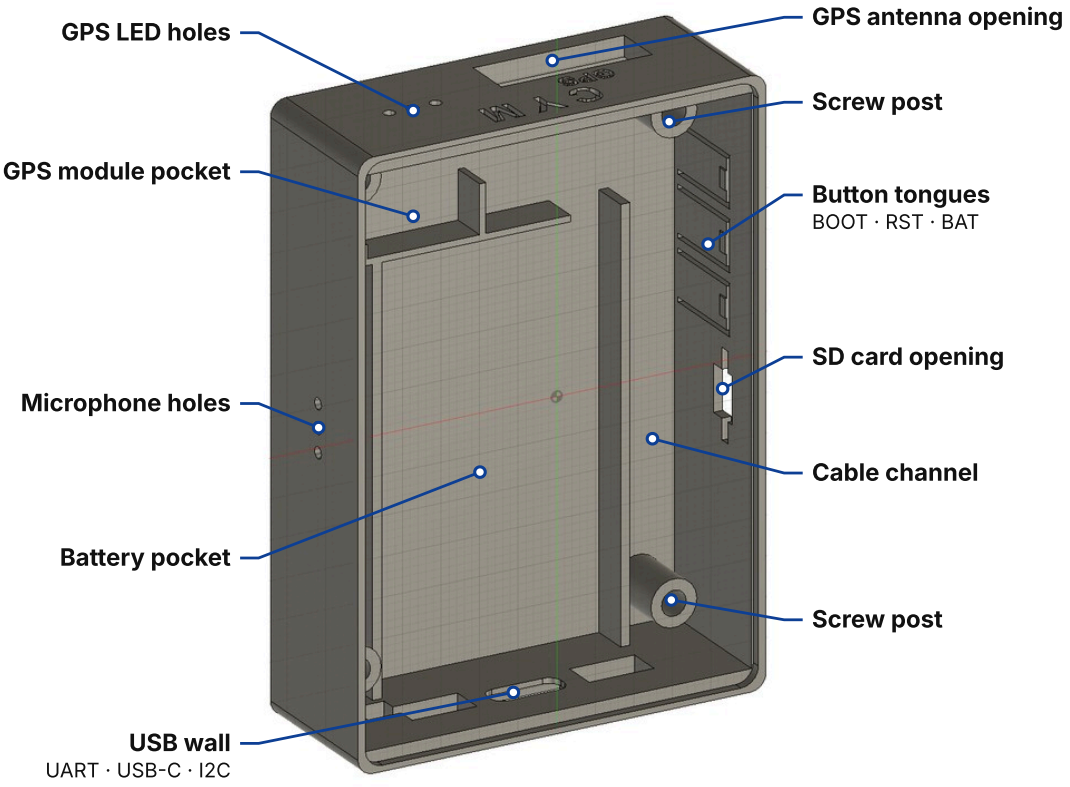
- Are the four screw holes and counterbores open?
- Do the three button tongues flex freely? If a slit is closed, open it with a craft knife.
- Are the USB, UART, I2C, SD and antenna openings clean?
- Are the microphone and GPS LED holes (1.5 mm) open? If not, open them by hand with a 1.5 mm drill bit.
- Are there burrs on the inner edge of the display frame, or strings inside the engraved lettering?

Part 3 – Assembly

What you need

- Printed case (CYM Case WS-C5-28)
- Waveshare ESP32-C5-Touch-LCD-2.8 board
- LiPo battery 1100 mAh, 53 × 30 × 7.5 mm, with 1.25 mm plug
- GPS module 16 × 14 × 5.5 mm with its antenna (19 × 6 × 5.5 mm)
- 4-wire cable with UART plug (wire outer diameter max. 1.3 mm)
- 4 × M2.5 × 16 cheese-head screws (head diameter max. 4.7 mm)
- Double-sided tape, adhesive tape, hot glue, soldering kit, tweezers

Interior layout

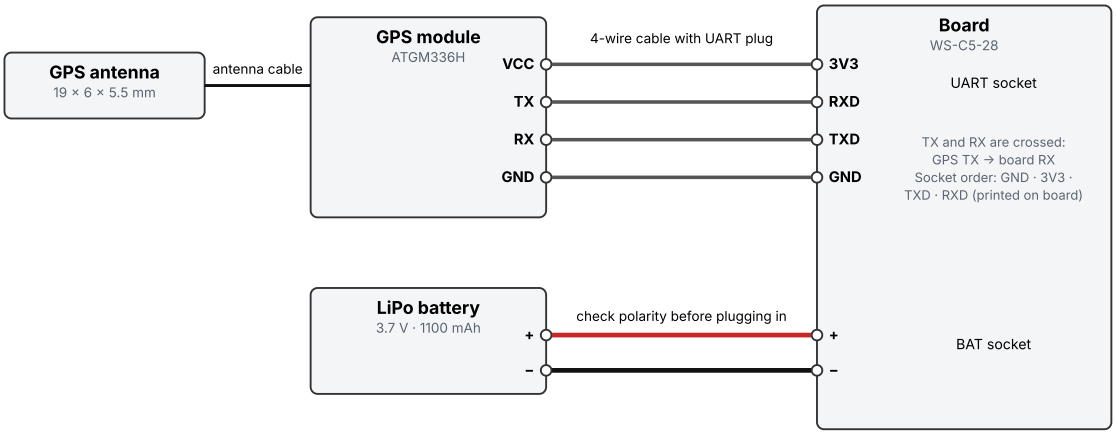


Inside of the case (viewed from the open side); USB wall at the bottom.

Component layout inside the case (open-side view, top end up).

Wiring diagram

Which wire goes where: the battery connects to the BAT socket of the board, the GPS module to the UART socket, and the antenna to the GPS module.



Verify the polarity with a multimeter before plugging in the battery.

> Caution: wire colors vary from set to set. Make the connections by socket and pad names, not by wire color (GPS VCC → 3V3, TX → RXD, RX → TXD, GND → GND on the board's UART socket; TX and RX are crossed), and verify the positive lead of the battery with a multimeter.

Assembly steps

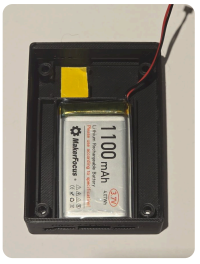
Step 1 Apply double-sided tape

Stick double-sided tape where the battery and the GPS module will sit: three pieces on the floor of the battery pocket, one piece in the GPS pocket in the top-left corner.



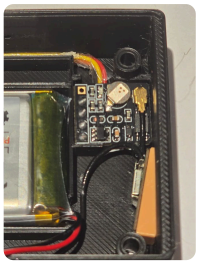
Step 2 Place the battery

Press the battery into its pocket against the USB wall, with its cable facing the GPS side (up). The cable leaves through the gap at the far right of the pocket's top edge. Do not plug the battery in yet.



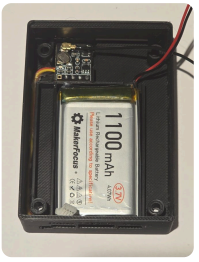
Step 3 Solder the GPS wires, connect the antenna

Solder the four wires directly to the pads of the GPS module; a pin header does not fit. Route the wires flat along the module, not upward. Then press the plug of the antenna cable onto the gold-colored socket on the module.



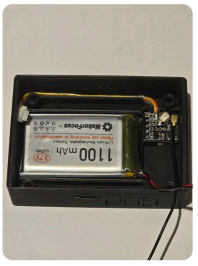
Step 4 Place the GPS module

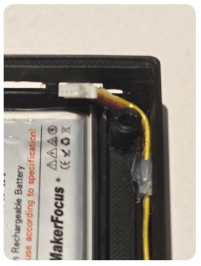
Press the module onto the tape in its pocket in the top-left corner, resting against the top wall. The pocket has only 0.3 mm of clearance lengthwise; if it is tight, lightly sand the module edge.



Step 5 Flatten the cable and run it through the channel

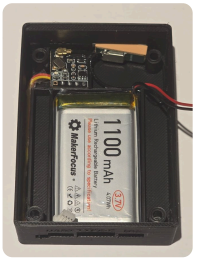
Lay the four wires side by side in a single row and wrap them with adhesive tape so the cable stays flat. Place the flat cable in the channel along the side wall and bring the plug down to the USB wall.

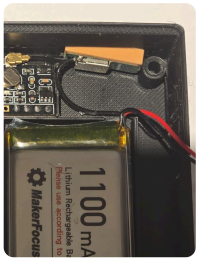




Step 6 Place the antenna

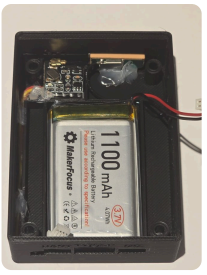
The antenna cable is long: gather the excess in a loop between the GPS module and the top-right screw post. Place the antenna in its opening in the top wall.





Step 7 Secure with hot glue

Secure the antenna, the antenna cable and the cable in the channel with a moderate amount of hot glue. Do not put glue in front of the three microphone holes in the middle of the long side wall or the three GPS LED holes in the top wall; it would close them.





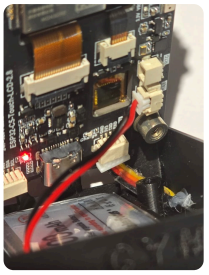
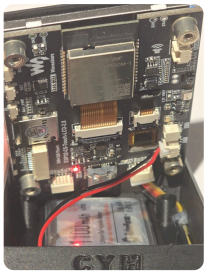
Step 8 Plug the GPS cable into the UART socket

Hold the board tilted above the case and plug the GPS cable into the UART socket on the board. The wires pass through the side pass-through on the corner side of the UART opening, not below the socket.



Step 9 Plug in the battery

After checking the polarity, plug the battery into the BAT socket on the board. The red LED lights up. Lay the excess cable flat on top of the battery.

Step 10 Seat the board in the case

Lower the board into the case with the display facing up; make sure no wires get pinched between the board and the case. The glass sits inside the frame, flush with the edge.



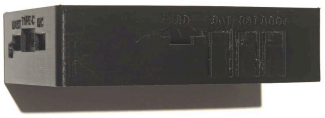
Step 11 Fit the screws

Turn the case over and fit the M2.5 × 16 screws through the four holes in the back into the board's own standoffs. Do not overtighten.



Step 12 Insert the SD card

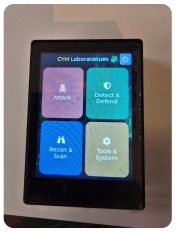
Insert the SD card last, through the slot in the button wall. The card sits flush with the wall. To remove it, grip the card from above and below with tweezers or a fingernail through the notch in the middle of the slot.



Step 13 **Final check**

The USB-C plug must enter its recess easily, the UART wires must come out through the side pass-through, the three buttons must click and the display must be flush with the frame. Switch the device on with the BAT button.





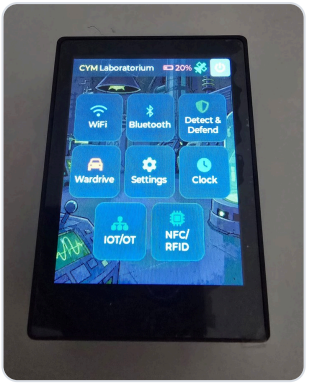
Part 4 – Firmware

Flashing the firmware with the Web Flasher

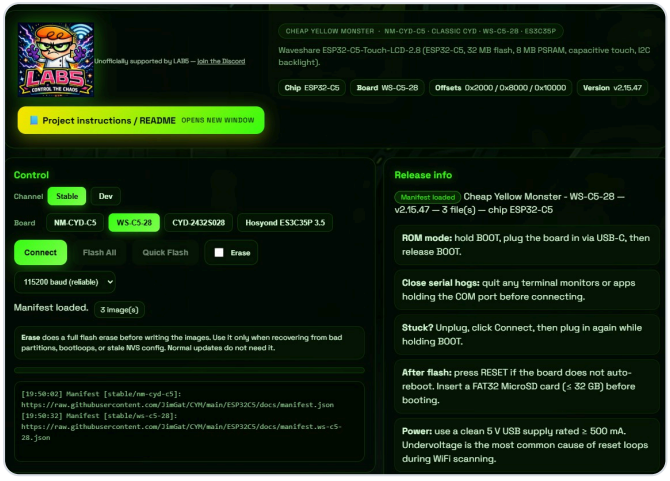
You can flash the firmware directly from the browser, without a USB driver or an IDE.

URL: https://jimgat.github.io/CYM/

1. Open jimgat.github.io/CYM/ in Chrome or Edge.
2. Select the "WS-C5-28" board.
3. Connect the board to the computer via the USB-C socket and click Connect.
4. Click the "Flash" button and wait for the process to complete.
5. The board restarts automatically; the CYM interface appears on the screen.



When flashing is complete, the CYM main menu appears on the screen.



Web Flasher interface – WS-C5-28 board selected.

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- License text: creativecommons.org/licenses/by-nc-nd/4.0
- The CYM firmware is a separate project under its own license. Board and part names are trademarks of their manufacturers.

> The Web Flasher works only in Chrome and Edge (requires the Web Serial API). Safari and Firefox are not supported.